THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The 'powerhouse of the cell' is the: 1

(a) ribosome **(b)** lysosome **(c)** Golgi apparatus **(d)** mitochondrion

2. Protein synthesis takes place on: 1

(a) ribosomes **(b)** lysosomes **(c)** vacuoles **(d)** Golgi bodies

3. Mitochondria and plastids resemble bacteria because they have their own: 1

(a) DNA and ribosomes **(b)** cell wall

(c) cell membrane only **(d)** nucleus

4. Growth of an organism is mainly due to: 1

(a) death of cells **(b)** decrease in cell size

(c) increase in cell size only **(d)** increase in the number of cells

5. Uncontrolled cell division that forms tumours is characteristic of: 1

(a) red blood cells **(b)** gametes **(c)** cancer cells **(d)** root hair cells

6. Epithelial tissue forms: 1

(a) the framework of the body **(b)** contractile fibres

(c) the outer covering and lining of organs **(d)** neurons

7. The tissue that provides cushioning at joints and in the nose and ear is: 1

(a) adipose **(b)** cartilage **(c)** blood **(d)** bone

8. Muscles that move food in the intestine are: 1

(a) tendons **(b)** skeletal muscles **(c)** cardiac muscles **(d)** smooth muscles

9. The joint between the skull and the first vertebra that allows turning the head side to side is a: 1

(a) pivot joint **(b)** fixed joint

(c) hinge joint **(d)** ball and socket joint

10. F. C. Steward grew whole carrot plants from single: 1

(a) root hair cells **(b)** xylem cells **(c)** guard cells **(d)** phloem cells

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name three types of plastids and their functions. 2

12. What is the difference between mitosis and meiosis in one line? 2

13. What is a joint? Name two types. 2

14. State the function of the rib cage. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. State any three postulates of the cell theory and its significance. 3

16. Why do dancers need flexible joints? Name two joints used in folk dance. 3

17. Name the organelles that synthesise the cell membrane and trace the path of its materials. 3

OR

Why does the carrot in plain water stay stiff but the one in strong salt solution becomes limp? 3

18. How does the rib cage help in breathing? 3

OR

Describe F. C. Steward's experiment on carrot phloem cells. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) How does the structure of a red blood cell suit its function? 5
(ii) Give another example of a specialised cell.
(iii) What is the length of an RBC lifespan?

OR

Case study: A student wants to estimate the size of cells under the microscope: a field of 4 mm shows 20 cells in a row; eyepiece 10X and objective 10X. 5
(i) Find the size of one cell.
(ii) What is the total magnification?
(iii) How big does one cell appear?
(iv) Which part of the microscope gives clarity?

20. (i) What is totipotency? 5
(ii) Describe Steward's carrot experiment.
(iii) Can animal cells be cultured into a whole animal in the same way?

OR

Case study: Aamrapali compares cuttings of a mango sapling. 5
(i) Which tissue lets the sapling bend?
(ii) Which tissue forms new roots on a cutting?
(iii) Name the tissue carrying water in the cutting.
(iv) State two parameters to keep the same in a comparison experiment.

END OF QUESTION PAPER — Total: 40 Marks

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